
Chapter 7

Foreign Currency Derivatives

Futures & Options

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Learning Objectives

01



How FX futures are quoted, valued, and used for speculation

02



Buying & writing FX options: risks and returns for each party

03



How option values change with exchange rate movements and time

04



How option values respond to price component changes (The Greeks)

What Are Derivatives?

A derivative is a financial instrument whose VALUE is derived from an underlying asset (e.g. a currency, commodity, or index).

Speculation

Take a position expecting to profit from price movements.
Example: Bet that the euro will rise vs. USD — buy a call option.

Hedging

Reduce risk from normal business operations.
Example: A Vietnamese exporter locks in a USD sell price using futures.

Key Benefits: Achieve payoffs not otherwise possible • Hedge unhedgeable risks • Reduce earnings volatility • Tax efficiency

Futures vs. Forward Contracts

Feature	Futures	Forward
Contract Size	Standardized (e.g. MXN 500,000)	Customizable
Maturity	Fixed dates (3rd Wed of month)	Any agreed date
Trading Venue	Organized exchange (CME/IMM)	OTC – bank & client
Margin	Initial margin + daily mark-to-market	Bank relationship only
Settlement	Rarely delivered (~2%)	Usually delivered (~95%)
Counterparty Risk	Exchange clearinghouse	Counterparty bank



Example: A US importer buys MXN 500,000 futures at \$0.055/MXN to lock in costs — exchange guarantees delivery.

Key Option Terms

Strike / Exercise Price

The rate at which the option holder can buy (call) or sell (put) the currency.

💡 e.g. Strike = \$1.10/€ — holder can buy euros at \$1.10 no matter the market rate.

Premium

The upfront cost paid by the buyer to the writer for the option contract.

💡 e.g. Premium = \$0.02/€ — buying a 1M € call costs \$20,000.

Holder vs Writer

Holder = buyer (pays premium, has the right). Writer/Grantor = seller (receives premium, has the obligation).

💡 Writer's max gain = premium. Holder's max loss = premium.

American vs European

American: can exercise any time before expiry. European: only on expiry date.

💡 Most exchange-traded FX options are American style.

FX Options – The Basics

An FX option gives the buyer the **RIGHT** (not obligation) to buy or sell currency at a fixed price by a set date.



CALL Option

Right: RIGHT to BUY foreign currency

Exercise when: Exercise when spot rises above strike price

Max Profit: Unlimited upside

Max Loss: Premium paid



PUT Option

Right: RIGHT to SELL foreign currency

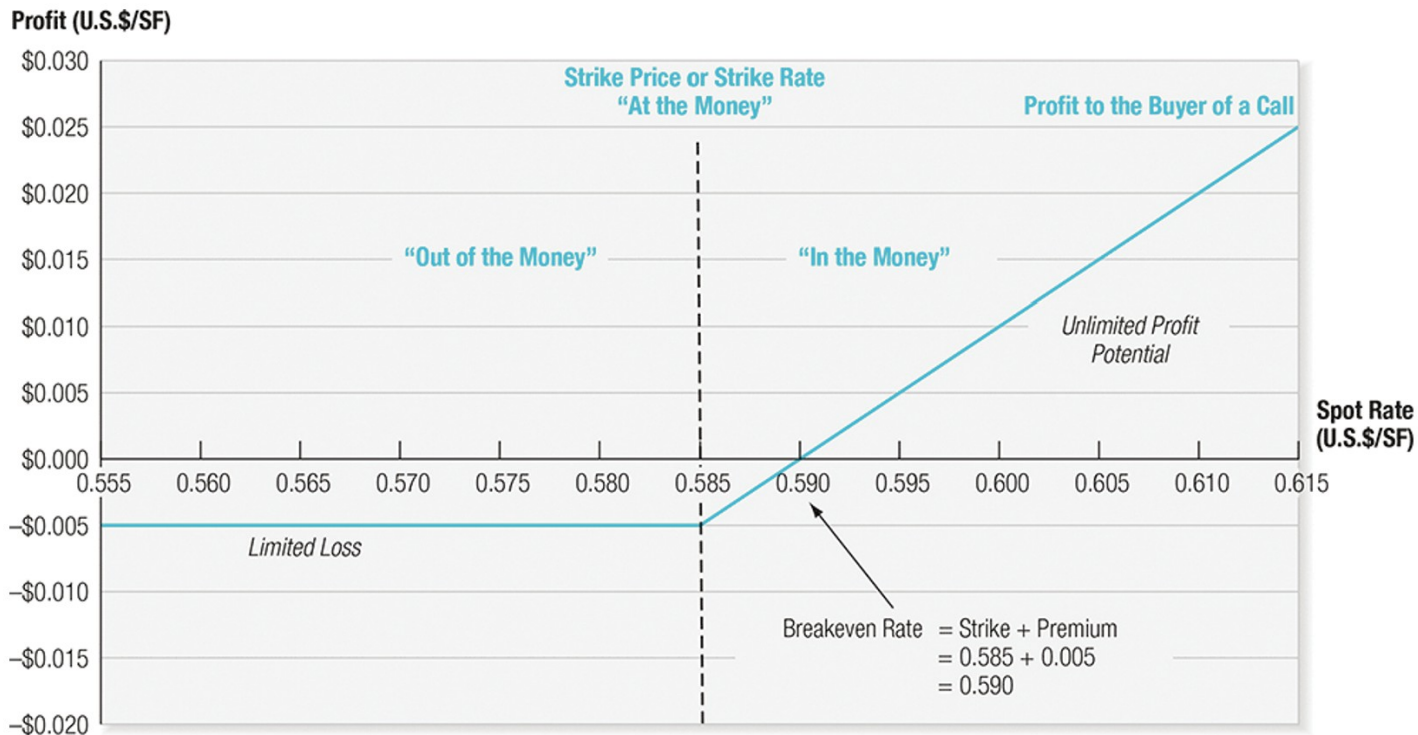
Exercise when: Exercise when spot falls below strike price

Max Profit: Large downside captured

Max Loss: Premium paid



Exhibit 7.3 Profit and Loss for the Buyer of a Call Option



The buyer of a call option has unlimited profit potential (*in the money*), and limited loss potential, the amount of the premium (*out of the money*).

Option Moneyness: ATM / ITM / OTM



ATM

AT-THE-MONEY

Condition:

Spot = Strike

Option has no intrinsic value — only time value.



ITM

IN-THE-MONEY

Condition:

Call: Spot > Strike

Put: Spot < Strike

Profitable to exercise immediately. Has intrinsic value.



OTM

OUT-OF-THE-MONEY

Condition:

Call: Spot < Strike

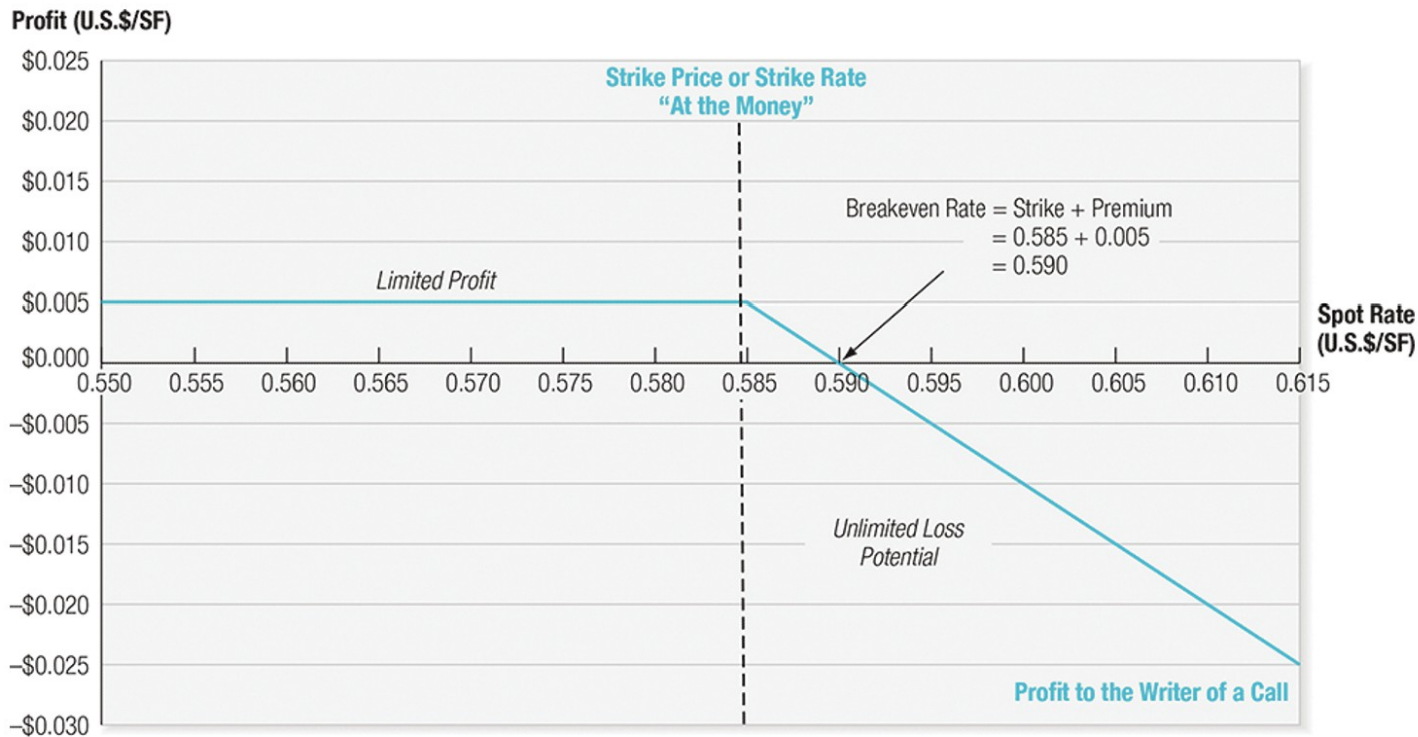
Put: Spot > Strike

Not profitable if exercised now. Only time value remains.

Example: You hold a CALL to buy CHF at \$0.585. If spot = \$0.610 → ITM (gain \$0.025/CHF). If spot = \$0.560 → OTM.



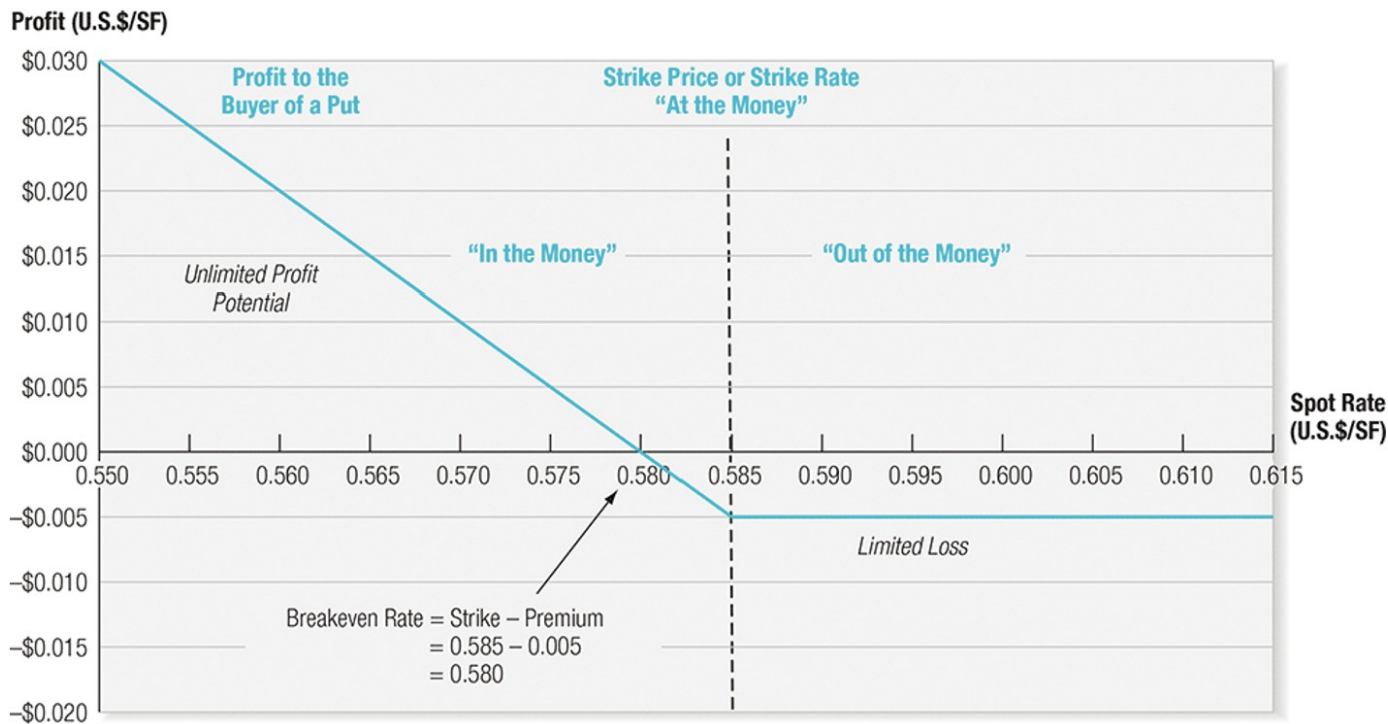
Exhibit 7.4 Profit and Loss for the Writer of a Call Option



The **writer of a call option** on has unlimited loss potential and limited profit potential, the amount of the premium.



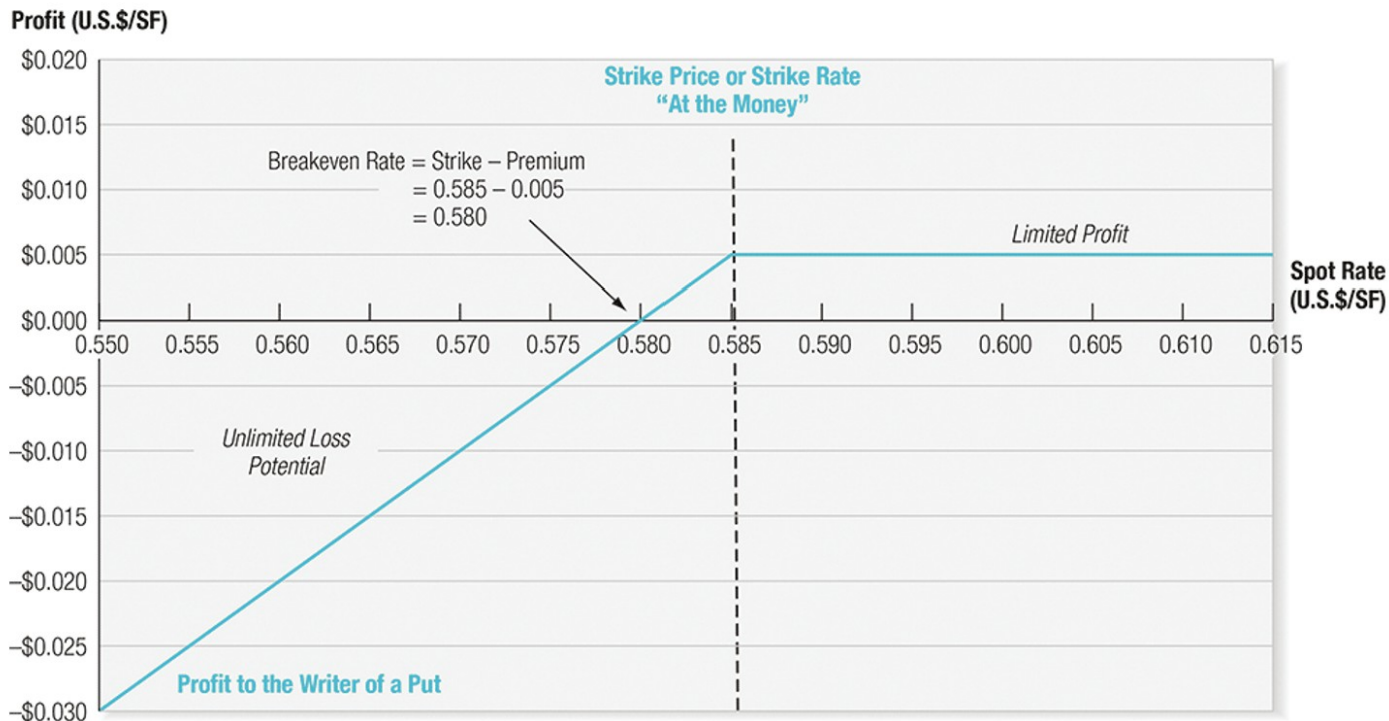
Exhibit 7.5 Profit and Loss for the Buyer of a Put Option



The **buyer of a put option** has unlimited profit potential (*in the money*), and limited loss potential, the amount of the premium (*out of the money*).



Exhibit 7.6 Profit and Loss for the Writer of a Put Option



The [writer of a put option](#) has limited profit potential, the premium, and an unlimited loss potential.

Option Pricing & Valuation

$$\text{Option Premium} = \text{Intrinsic Value} + \text{Time Value}$$

Intrinsic Value

The gain if the option were exercised RIGHT NOW.

Call: $\max(\text{Spot} - \text{Strike}, 0)$

Put: $\max(\text{Strike} - \text{Spot}, 0)$

ITM option → positive intrinsic value

ATM/OTM option → intrinsic value = ZERO

Time Value

Extra premium above intrinsic value.

Reflects the possibility that the spot moves further ITM before expiry.

Declines as expiry approaches (theta decay)

At expiry: Time Value = 0

Longer maturity → larger time value

6 Inputs That Determine the Premium:

1. Spot Rate

2. Time to Maturity

3. Forward Rate

4. USD Interest Rate

5. Foreign Interest Rate

6. Volatility (σ)



Exhibit 7.7 Option Intrinsic Value, Time Value, and Total Value

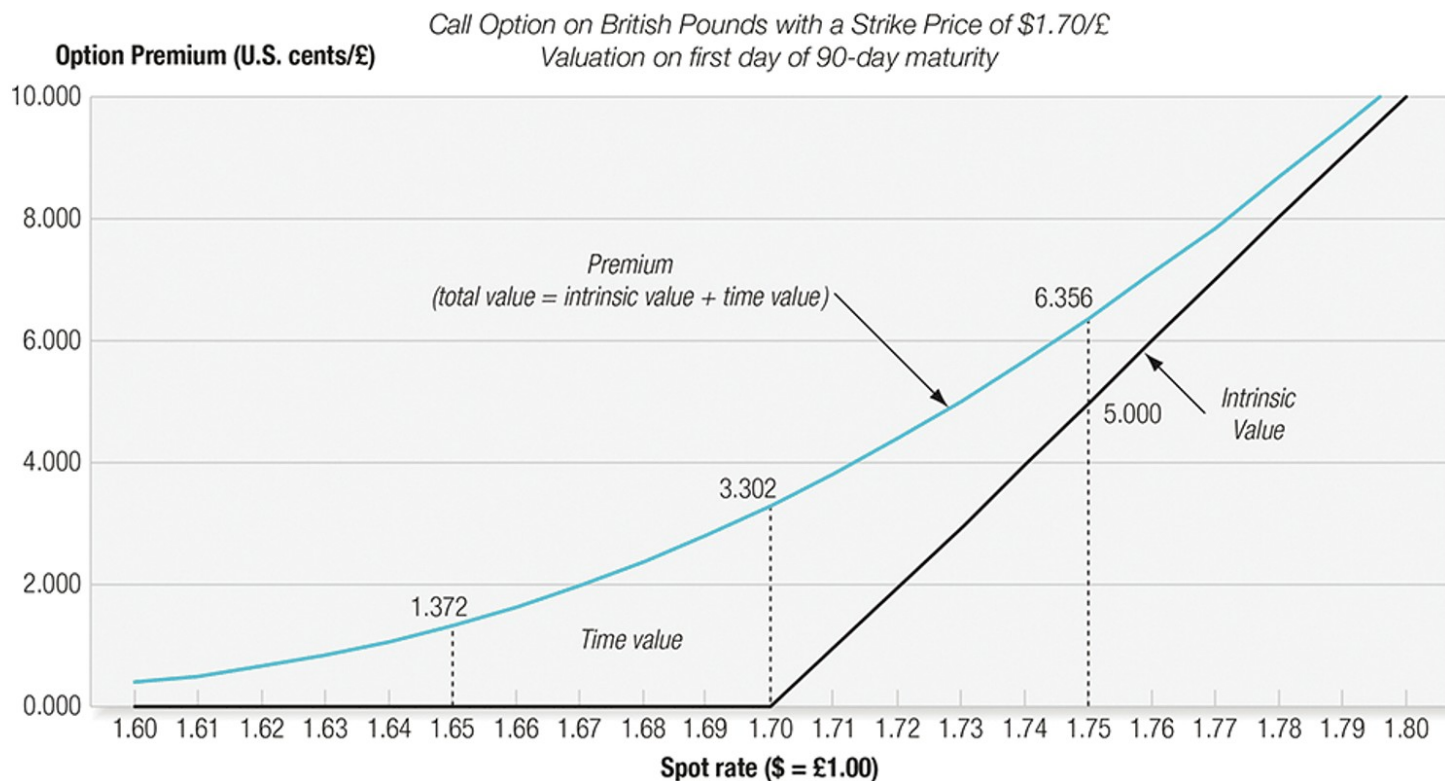




Exhibit 7.8 Call Option Premiums: Intrinsic Value and Time Value Components

Strike Rate (\$/£)	Spot Rate (\$/£)	Money	Call Premium (U.S. cents/£)	=	Intrinsic Value (U.S. cents/£)	+	Time Value (U.S. cents/£)	Option Delta (0 to 1)
1.70	1.75	In-the-money (ITM)	6.37	=	5.00	+	1.37	0.71
1.70	1.70	At-the-money (ATM)	3.30	=	0.00	+	3.30	0.50
1.70	1.65	Out-of-the-money (OTM)	1.37	=	0.00	+	1.37	0.28



Currency Option Pricing Sensitivity

- If currency options are to be used effectively, either for the purposes of speculation or risk management, the individual trader needs to know how option values (premiums) react to their various components.
- Six sensitivities:
 1. The impact of changing forward rates
 2. The impact of changing spot rates
 3. The impact of time to maturity
 4. The impact of changing volatility
 5. The impact of changing interest differentials
 6. The impact of alternative option strike prices

Sensitivity	Sign / Range	Interpretation	Key Trading Insight
Forward Rate Sensitivity	Depends on position	Measures how option value changes as the forward FX rate changes.	Since FX option pricing incorporates spot rates and both domestic and foreign interest rates, the forward rate is the central valuation benchmark.
Spot Rate Sensitivity <i>Delta (δ)</i>	Call: 0 to +1 Put: -1 to 0	It tells you how much the option price changes when the spot exchange rate changes.	Higher delta $\Rightarrow$ higher probability of expiring ITM. Delta > 0.7 usually indicates a deep ITM call.
Time Sensitivity <i>Theta (θ)</i>	negative	It tells you how much value disappears as time passes.	Premiums decay faster as expiration approaches. Longer-dated options retain value better.
Volatility Sensitivity <i>Lambda (λ)</i>	positive	Measures the impact of changes in expected volatility on option value. Higher volatility increases the probability of large favorable moves.	Higher volatility $\Rightarrow$ higher option premium. Traders expecting volatility to fall may sell options.
Domestic Interest Rate Sensitivity <i>Rho (ρ)</i>	Generally positive for calls	Measures how option value changes when domestic interest rates change.	Buy currency calls before domestic rates rise because call premiums tend to increase.
Foreign Interest Rate Sensitivity <i>Phi (ϕ)</i>	Generally negative for calls	Measures how option value changes when foreign interest rates change.	Higher foreign interest rates tend to reduce call premiums due to the FX cost-of-carry effect.
Strike Price Sensitivity	Varies by moneyness	How the option value changes when you choose a different strike price.	Lower strike calls (or higher strike puts) have higher intrinsic value and are generally more expensive.



Practice Questions

Q1

A US company buys a Swiss Franc CALL option: Strike \$0.58, Premium \$0.04. At expiry, spot = \$0.65. What is the net profit per franc?

Q2

You WRITE a put option on euros: Strike \$1.10, Premium \$0.03. At expiry, spot = \$1.05. What is your P&L?

Q3

A CHF call option has Strike = \$0.60 and current spot = \$0.55. Is this option ITM, OTM, or ATM? What is its intrinsic value?

Q4

Option delta = 0.80. What does this tell you about the probability of expiring ITM? What happens to delta if the option moves deeper ITM?

Real-World Applications



Vietnamese Exporter (Hedger)

Situation: A Ho Chi Minh City factory exports \$2M of goods to the US, to be paid in 3 months.

Strategy: ?

Outcome: ?



Speculator – Betting on Euro Strength

Situation: A trader believes the euro will strengthen from \$1.08 to \$1.15 in 60 days.

Strategy: ?

Outcome: ?



Japanese Firm – Income from Writing Puts

Situation: A Japanese firm is comfortable buying USD at ¥148 or better.

Strategy: ?

Outcome: ?

Chapter 7 – Key Takeaways



Futures vs. Forwards

Futures are standardized, exchange-traded, rarely settled. Forwards are customized, OTC, usually settled.



Moneyness

ITM = profitable to exercise now. ATM = break-even. OTM = not profitable.



The Greeks

Delta (spot), Theta (time), Lambda (vol), Rho (domestic rate), Phi (foreign rate) measure sensitivity of premiums.



Calls & Puts

Call = right to buy. Put = right to sell. Buyer's max loss = premium. Writer's risk can be unlimited.



Option Pricing

Premium = Intrinsic Value + Time Value. Determined by 6 factors including spot rate, volatility, and interest rates.



Applications

Used by MNEs to hedge FX exposure, and by traders to speculate on currency movements with limited downside.